

Final Design Report

Capstone F1 - High Altitude Platforms
RF Communications

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1 Introduction

Reliable connectivity in rural and remote regions remains difficult to achieve. Long-fibre installations and dense terrestrial towers are often very expensive in sparsely populated areas and on demanding terrain [1]. In response to this gap, this project investigates whether a balloon-based high-altitude platform (HAP) surrogate can provide a practical, low-bandwidth, bi-directional communication link to ground users. Working with the client, the team focused on maximizing line-of-sight communication performance. They also kept payload weight, power consumption, and cost within realistic limits for student-built hardware.

This direction is supported by research on aerial communication systems. Studies of HAPs show that stratospheric platforms can cost-effectively extend wireless coverage. They may serve as radio-access or backhaul nodes for regions underserved by terrestrial towers [2]. Unmanned Aerial Vehicle (UAV) communication research similarly notes that starting with single airborne nodes is effective. These nodes exploit strong line-of-sight channels and avoid the added complexity of multi-platform coordination [3]. For these reasons, the project centred on a single airborne RF link as a suitable proof of concept. More complex multi-platform or hybrid RF and optical systems were treated as outside the scope of this capstone project.

The broader industry context reinforces the relevance of this approach. International Telecommunication Union (ITU) statistics show that billions remain offline or under-connected due to the high cost of terrestrial infrastructure [1]. Aerial platforms offer an intermediate solution, combining extensive coverage with lower deployment cost than full ground networks and lower latency than satellite-only approaches [2]. Low-power wide-area network (LPWAN) technologies such as LoRa and NB-IoT already demonstrate multi-kilometre rural coverage at low power, confirming that long-range, low-bit-rate links are feasible for applications similar to this project [4]. Balloon-based experiments at 868 MHz and 2.4 GHz also show that these ISM bands remain usable in near-space environments and provide data that can inform antenna and link-budget decisions [5].

Within this context, the F1 project developed a modular RF communications payload intended as a realistic HAP-link demonstrator. The final prototype is centred on a Raspberry Pi Zero 2W host system and integrates multiple communication technologies within a single architecture, including Wi-Fi HaLow for moderate-range, higher-throughput communication and LoRa for long-range, low-power communication, while 4G/LTE remains part of the planned system expansion. The resulting prototype demonstrates the feasibility of a modular extended-range communication platform within student-level constraints, while also revealing remaining integration challenges that inform the final evaluation and future work sections of this report.

2 Problem Scope and Constraints

The problem scope was refined through client feedback and background research to define a realistic proof of concept for extended-range communications using a HAP surrogate. Although the original problem allowed several possible deployment contexts and technology directions, the final scope was narrowed to a low-bandwidth RF communications subsystem that could be implemented, tested, and justified within the time, cost, and technical limits of a capstone project.

The scope of this project included:

- development of a proof-of-concept bidirectional wireless communications link;
- support for essential low-bandwidth communication, such as text messaging and short voice-related data;
- operation in licence-exempt spectrum subject to relevant ISSED requirements;
- implementation on a modular payload suitable for low-altitude or ground-accessible testing;
- design decisions consistent with a balloon-based HAP use-case, even where final testing used surrogate deployment conditions.

The following items were outside the scope of the final prototype:

- full autonomous HAP deployment and flight operations;

- a complete cellular-standard base-station implementation;
- fully integrated multi-node mesh networking;
- complete 4G/LTE backhaul implementation;
- full co-optimization of the RF payload with the mechanical HAPS platform.

This project remained independent from the other HAPS capstone teams, although coordination with them could improve future integration. In particular, closer coordination with the mechanical platform team would improve payload packaging, antenna placement, cable routing, mounting stability, and overall deployment readiness.

2.1 Stakeholders

The primary stakeholder is Dr. Anas Chaaban, the client for the HAPS projects at UBC Okanagan. The project addresses the need for sovereign communications capability in remote or underserved regions, especially where conventional terrestrial infrastructure is costly or impractical. Other stakeholders include residents of underserved areas and emergency-response personnel. Future development teams, telecommunications organizations, and the broader public also play a role, particularly in situations involving communication resilience during disasters or remote operations. These stakeholders influence project needs for practical deployment, reliable low-bandwidth operation, manageable payload size, and safe operation.

2.2 Technologies

RF communications and spectrum utilization are governed by ISED technical standards, and any implementation must comply with the relevant Radio Standards Specifications (RSS). Due to time and resource constraints, this project was limited to the use of licence-exempt spectrum. RSS-247 [6] outlines requirements for operation in licence-exempt bands relevant to technologies such as Wi-Fi and LoRa. Relevant technologies considered within this project included LoRa, Wi-Fi HaLow, and related low-power wireless approaches appropriate for extended-range, low-bandwidth communication. Relevant technologies under license-exempt use include Wi-Fi, Bluetooth LE, LoRa, and WiMAX.

2.3 Existing Solutions

Existing remote communication solutions do not fully address the intended use case of this project. Satellite communication can serve regions without cellular coverage, but it may be costly, may have limited accessibility, and may depend on foreign infrastructure. Large-scale HAP systems and LTE/5G aerial repeaters can provide wide-area coverage, but their complexity and cost are not well matched to a student-built proof-of-concept. Tethered balloon and aerostat platforms are more physically relevant but typically do not provide an accessible, ready-made, low-cost communication payload for the use case considered here.

3 Design Specifications

The design specifications define the benchmark for evaluating the final prototype. These specifications come from the project scope, stakeholder needs, and limitations of existing solutions. In the final report, they serve two purposes. First, they are design targets guiding development. Second, they are benchmarks for assessing final design performance and project success. The final design followed an extended-range proof-of-concept direction. It prioritized communication range while keeping payload mass, power consumption, cost, and low-bandwidth functionality acceptable. Following client feedback, throughput was mainly a threshold requirement. Weight, power, range, and reliability stayed as the main evaluation priorities.

3.1 Design Requirements

The design requirements are mandatory conditions that the prototype must satisfy to be considered successful.

Table 1: Design requirements for the HAP communication system

ID	Requirement	Target	Reason
R1	Bidirectional communication link	Demonstrate a working two-way wireless link between the HAP node and at least one ground device	Confirms the system provides usable connectivity.
R2	Minimum communication capability	Support low-bandwidth communication, with at least text-based messaging and a minimum data rate of about 2 kbps	Matches the emergency-oriented use-case.
R3	Battery operation	Operate from an onboard battery for at least 30 minutes during demonstration conditions	Reflects limited-power deployment conditions
R4	Demonstration platform	Be demonstrable on a low-altitude surrogate platform or equivalent ground-accessible setup	Aligns evaluation with the agreed capstone scope.
R5	Payload mass	Maintain total payload mass below approximately 2 kg	Preserves practical deployability on a balloon-based platform.
R6	Cost	Maintain prototype cost within the project budget constraint of \$500, where feasible	Reflects project and procurement constraints.
R7	Regulatory compliance	Use licence-exempt spectrum in accordance with applicable ISED requirements	Ensures technically responsible implementation.
R8	Scope compliance	Remain within the agreed subsystem-level scope of the project	Prevents unjustified expansion into out-of-scope features.
R9	Balloon-based HAP	Design must use a tethered balloon as a platform; other platforms may be used for testing.	The client requested that the HAP be installed on a balloon.
R10	Regulatory compliance	Operate in a Canadian ISED-compliant licence-exempt band.	Must satisfy regulatory requirements.
R11	Extended communication range focus	Achieve ≥ 5 km line-of-sight in full-scale design; the scaled demo must show a meaningful advantage over local-only links.	The client emphasized the importance of range capabilities.

Table 2: Evaluation criteria for comparing feasible designs

Rank	Criteria	Target	Reason
E1	Payload mass and average power draw	Minimize (kg, W).	Top priority; directly drives deployability and mission runtime.
E2	Achievable line-of-sight communication range	Maximize (km).	A key differentiator of the selected design: matches the range-focused scope.
E3	Reliability and availability	Maximize (packet success percentage/uptime) within the target range.	Needs robust, The link should remain stable and usable under expected operating conditions, with low-loss messaging during deployment.
E4	Sustainability	Minimize (kg of CO ₂ or equivalent impact).	Encourages responsible material such as reusable parts, subsystem choices and waste handling (upcycling components).

3.2 Use of Specifications to Benchmark Solutions

These specifications are used as a reference for the subsequent sections on Final Design Performance Evaluation and Project Success Analysis. Each design requirement is assessed as met, partially met, or not met, based on the evidence from the prototype. Each evaluation criterion is then used to discuss the extent to which the final design achieved the intended trade-off between range, reliability, payload constraints, power use, and practicality.

4 Design Process and Final Solution Selection

The design process for the HAPS RF subsystem was carried out in a structured, methodical manner rather than selecting hardware opportunistically. The process was divided into two major stages. Firstly, candidate communication technologies and system concepts were scanned to identify the most technically credible baseline architecture. Secondly, that baseline was refined into a modular minimum viable prototype (MVP) and then iteratively improved to address the most important remaining limitations. Throughout both stages, the same engineering criteria were applied: achievable range, power consumption, payload mass, reliability, software and integration complexity, and overall cost. This ensured that the final solution was selected based on total mission fit rather than a single attractive feature.

4.1 Stage 1: Baseline Architecture Down-Selection

The first stage of the design process focused on identifying the most suitable foundational communication approach for a high-altitude platform. At the protocol level, the team considered LoRa, WiFi HaLow, unlicensed LTE-style SDR implementations, and Sigfox. This comparison was governed by core RF engineering principles. In particular, sub-GHz operation was attractive because it reduces free-space path loss relative to higher-frequency links and is therefore better suited to long-range line-of-sight communication. In addition, the choice of modulation strongly affected the trade-off between data rate and receiver sensitivity. LoRa, which uses chirp spread spectrum, offered a particularly strong balance for this project because it prioritizes sensitivity, robustness, and long-range operation over raw throughput. By contrast, higher-throughput approaches offered greater bandwidth, but at the cost of increased power draw, heavier processing requirements, and substantially higher implementation complexity.

These principles led to the comparison of three system-level concepts in the early design phase: a lightweight point-to-point LoRa link, an extended-range LoRa gateway architecture, and an

SDR-based LTE-style link. The results of that comparison are summarized in Table 3.

Concept	Cost (CAD)	Estimated Payload Mass	Power / Runtime	Technical Interpretation	Score
LoRa point-to-point	208.63	141.1-151.6 g	0.045 W steady-state, 0.50 W peak, 59-296 h depending on duty cycle	Very efficient in mass and power, but offers less range-focused reception capability and lower measurement flexibility than a gateway-based LoRa architecture.	4.15
Extended-range LoRa gateway	208.63	141.1-151.6 g	Same low-power envelope as the LoRa point-to-point solution	Preserves the long-range and low-power strengths of LoRa while improving reception probability, logging capability, and range characterization.	4.90
SDR LTE-style link	1007.29	421-581 g	About 6.2 W active, about 2.4-3.0 h runtime	Offers flexibility and higher-throughput experimentation, but exceeds the project constraints in cost, airborne power burden, payload mass, and software complexity.	1.95

Table 3: Early-Stage Concept Comparison Used to Select the Baseline RF Architecture.

From this first-stage screening, the LoRa-based architecture was selected as the correct foundation for this project. This decision was justified by several factors. Firstly, LoRa is directly aligned with the client’s extended-range focus while keeping airborne power and payload mass low. Secondly, the LoRa-based concepts remained within the project budget, whereas the SDR approach increased costs by nearly fivefold, with a severe penalty in runtime and integration risk. Thirdly, the LoRa gateway concept outperformed the simpler LoRa point-to-point approach by improving practical range characterization and reception reliability without sacrificing the low-power advantages of the LoRa physical layer. Therefore, the design process did not simply conclude that LoRa was good enough; it concluded, more specifically, that a LoRa-centred architecture was the best technical foundation for the final HAPS system.

4.2 Stage 2: Modular MVP Development

Once the foundational communication method has been selected, the next design step is to embody that decision in a hardware architecture that could support future growth. The resulting MVP was intentionally modular. The HAP node was reconstructed around a Raspberry Pi Zero 2 W single-board computer connected to radio modules over USB or SPI, while the base communication link was implemented through an ESP32-S3/SX1262 LoRa combination. This was an important design decision because it separated high-level networking logic from the underlying physical radio, thereby reducing redesign effort when evaluating later upgrades. The inclusion of Reticulum compatibility was also deliberate: it created a technology-agnostic network layer that could integrate multiple communication interfaces into a cohesive system. Likewise, the selected power management hardware was sized to support solar charging, USB-PD charging, and future expansion to additional radios. In other words, the MVP was not treated as a disposable prototype, but rather designed as a scalable platform for controlled iteration.

4.3 Stage 3: Iterative Improvement of the Selected Baseline

After establishing the modular LoRa-based MVP, the design process entered an iterative phase in which three viable upgrades were explored. Each iteration targeted a specific remaining weakness in the system.

- Iteration 1 added a 4G/LTE modem to address the fact that the LoRa MVP operated essentially as an isolated local area network. From a systems perspective, this was a network architecture problem rather than a radio link problem. Adding an LTE module transformed the HAPS module into a gateway that routes traffic between the local HAPS network and external wide-area networks. This was a significant improvement because it addressed one of the most important functional limitations of the MVP without replacing the low-power, long-range LoRa access layer.

- Iteration 2 WiFi HaLow to address the throughput limitation of LoRa. This iteration was based on sound communication-systems reasoning: LoRa is excellent for robust low-bandwidth links, but it is not efficient for applications that require higher local throughput. WiFi HaLow, therefore, offered a complementary short-range, higher-bandwidth link while preserving LoRa as a fallback for long-range communication. However, this benefit came with increased protocol overhead, more demanding link-management logic, and higher power consumption.
- Iteration 3 combined both LTE and WiFi HaLow in a single architecture. In principle, this was the most functionally complete design because it offered LoRa for long range, WiFi HaLow for higher local throughput, and LTE for backhaul connectivity. In practice, however, it exposed the project’s complexity ceiling. It imposed the highest software burden, the highest power draw, and the greatest integration risk because routing, traffic prioritization, and link-selection logic had to be handled across multiple active communication interfaces.

The quantitative comparison of those three iterations is shown in Table 4. The cost and mass values are shown as additions to the MVP, whose base cost was \$313.72 CAD, and whose base payload mass was approximately 101.0 grams.

Iteration	Added Functionality	Added Cost (CAD)	Added Avg. Power	Added Mass	Runtime (2 cells)	Main Technical Drawback
4G/LTE	WAN backhaul and gateway capability	62.00	537 mW	64 g	15.8 h	Requires cellular coverage and adds routing, security, and network-management complexity.
WiFi HaLow	Higher local throughput for nearby users	76.84	330-450 mW	39 g	17.9 h	Improves bandwidth but does not solve network isolation. Also increases protocol overhead and link-management complexity.
LTE + HaLow	Both gateway connectivity and higher local throughput	138.84	867-987 mW	77 g	12.1 h	Highest overall power draw, control complexity, and integration burden.

Table 4: System-Level Comparison of the Main Design Iterations Built on the Modular LoRa MVP.

The comparison in Table 4 shows that all three iterations were technically viable but not equally appropriate for the current stage of the project. Importantly, range was not the strongest discriminator in this phase because all three options preserved the LoRa backbone. The link-budget analysis for 5 km line-of-sight operation showed a strong fade margin for LoRa and an acceptable positive margin for WiFi HaLow, which confirmed that the sub-GHz communication layer remained fundamentally sound. As a result, the more important differentiators became added system value, deployability, integration burden, and mission endurance.

Criterion	Weight	Engineering Justification
Payload Mass / Power Consumption	0.35	These criteria were assigned the highest weight because airborne payload mass and onboard power draw directly affect flight feasibility, mission endurance, and overall deployability of the HAPS platform. A communication system with excessive mass or power demand would reduce practical operating time and place greater strain on the balloon payload budget.
Achievable Range	0.35	Range was weighted equally to mass and power because extended communication distance is a primary project objective. Since the HAPS system is intended to provide long-range low-bandwidth connectivity, the selected design had to demonstrate strong link feasibility under line-of-sight operating conditions.
Reliability / Stability	0.20	Reliability was given a moderate weight because consistent communication performance is necessary for the system to be useful in practice. Even if a design performs well in theory, frequent link drops, unstable network behaviour, or difficult recovery would reduce its operational value.
Sustainability / Scalability	0.10	Sustainability was assigned a lower, but still meaningful, weight because the project is currently at the prototype stage. While future expandability, modularity, and long-term maintainability are important, they were treated as secondary to immediate technical feasibility and mission performance.
Total	1.00	The weights sum to 1.00 and were used to compare alternative designs in a structured and objective manner.

Table 5: Weighted Decision Matrix Criteria Used to Evaluate HAPS Communication System Design Alternatives.

Table 5 summarizes the weighted criteria used to evaluate the candidate system architectures and guide the final design selection.

4.4 Final Solution Selection

Based on the complete design process, the final selected solution was not treated as a single one-step hardware choice, but rather as a phased modular system architecture. The client-directed final path was to begin with a fully functional and validated LoRa minimum viable prototype (MVP), and then iteratively expand the platform by first integrating WiFi HaLow and subsequently LTE, with optional combined operation once the baseline system performance had been confirmed. This approach was selected to balance technical ambition with implementation risk, schedule constraints, and prototype reliability. This decision reflects an important outcome of the design process. From a purely weighted decision-matrix perspective, the LTE-only iteration achieved the highest standalone score because it provided the greatest immediate improvement in connectivity while maintaining reasonable cost, payload, and power trade-offs. However, the client expressed a clear preference for a more capable long-term system that includes both LTE and WiFi HaLow, provided that the LoRa MVP is first demonstrated to work reliably end-to-end. Therefore, the final solution selection was based not only on numerical scoring, but also on stakeholder priorities, modular feasibility, and risk-managed implementation sequencing. The selected phased architecture is technically justified by the complementary strengths of the three communication layers. LoRa remains the foundation of the system because it best satisfies the project’s core requirement of long-range, low-power communication. WiFi HaLow is introduced afterward to improve local throughput when devices are within useful range, while LoRa continues to serve as the fallback long-range link. LTE is then added to extend the system beyond a local HAPS network by providing wide-area backhaul capability and allowing the node to function as a gateway between the airborne platform and external networks. In combination, these protocols provide a more flexible and capable communication system than any single protocol alone. The phased integration strategy also demonstrates the structured application of engineering design principles. By requiring the LoRa MVP to be validated before adding new modules, the team reduces debugging ambiguity, more effectively isolates subsystem failures, and avoids introducing unnecessary complexity before the baseline link is proven. This is especially

important in a capstone prototype, where software integration effort, hardware compatibility, power budgeting, and testing time must all be managed carefully. The modular sequence can be shown as:

LoRa MVP → WiFi HaLow integration → LTE integration → optional combined operation

Therefore, it provides a practical path toward the client’s desired final prototype while preserving technical rigour and reducing project risk. Accordingly, the final solution for this project is best described as a modular multi-protocol HAPS communication platform built on a validated LoRa MVP, with planned LTE and WiFi HaLow expansion. This solution was selected because it most effectively satisfies the project objectives, aligns with the client’s direction, remains feasible within the remaining timeline, and provides the strongest overall balance among performance, flexibility, scalability, and implementation risk.

5 Final Design Description

This section provides a technical specification of the final High-Altitude Platform (HAP) communications payload. The design is engineered as a modular, multi-protocol gateway capable of providing resilient IP and mesh connectivity in remote environments.

5.1 System Architecture Overview

The HAP architecture utilizes a modular platform centered on a **Raspberry Pi Zero 2 W** single-board computer (SBC). The system is designed to function as a versatile communications gateway, bridging long-range Sub-1GHz links with cellular backhaul and local user hotspots.

The architecture integrates three primary radio interfaces to ensure connectivity across varying environments:

- **802.11ah (WiFi HaLow):** Provides long-range IP links (demonstrated up to 16 km by Morse Micro [7]) using the Morse Micro MM6108 chipset. This serves as the primary high-throughput Sub-1GHz interface for the HAP-to-Ground and HAP-to-Client links.
- **LTE Cat 1 bis:** Facilitates cellular backhaul for internet connectivity where signal is available, enabling the HAP to act as a wide-area network (WAN) gateway.
- **LoRa:** Utilized for ultra-low-bandwidth, long-range messaging and mesh transport via the Reticulum Network Stack.

The system operates on Raspberry Pi OS (Debian Trixie) and employs the **Reticulum Network Stack** as its core transport layer. This framework allows the HAP to act as a transport and propagation node, routing data packets seamlessly across diverse physical media, including serial, USB, and Ethernet interfaces.

5.2 Hardware Configuration and Physical Integration

The final prototype is constructed using low-cost COTS (Commercial Off-The-Shelf) components integrated into a custom 3D-printed enclosure. The final prototype weight is 591 g, and dimensions of 10 cm × 10 cm × 14.5 cm excluding antennas. Antennas measure 19.5 cm in straight configuration, with a bending joint at 2.5 cm distance from the connector.

An additional 3D printed enclosure was designed to house the HaLow client radio module. The LoRa radio kit includes a printed case.

5.2.1 Hardware Specification Table

Table 6 summarizes the technical specifications of the core components used in the final design.

5.2.2 Modular Physical Integration

The system utilizes a vertical stack configuration. The USB Hub HAT connects to the RPi Zero 2 W via pogo pins, eliminating the need for external micro-USB connection. LoRa radio modules are connected to MCUs following the XIAO form factor, allowing for component interchangeability. The HaLow radio connects directly to the SBC using SPI via jumper wires.

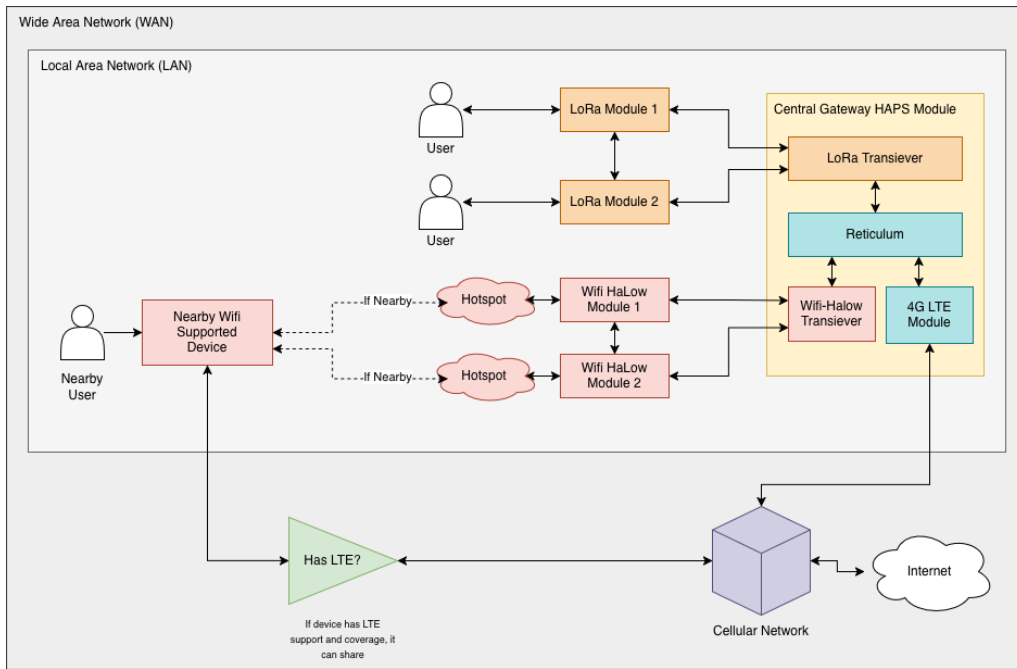


Figure 1: HAP System Network Architecture Diagram, illustrating connection between LTE backhaul, HaLow IP links, and the LoRa Reticulum transport layer.



Figure 2: Final HAP prototype with LoRa and HaLow client devices (right to left).

Table 6: Technical Specifications and Selection Rationale for HAP Gateway V1.0

Component	Specification	Selection Rationale
SBC	Raspberry Pi Zero 2 W	High availability and compatibility with expansion hardware via 40-pin GPIO/test pads.
LTE Modem	BKP-SIM7670G-MNGV (SIM7670G Cat 1 bis)	Global band support; supports USB-CDC/RNDIS for native IP addressing within Linux. LTE-only reduces peak current to 800 mA.
HaLow Radio	Seeed XIAO Quectel FGH100M-H (Morse Micro MM6108)	Provides 802.11ah compliance with 27 dBm output for long-range performance.
MCU (Radios)	Seeed XIAO ESP32-S3	Standardized form factor for radio modules; maintains a single firmware development ecosystem. Integrated Wi-Fi/BLE for client bridging.
LoRa Radio	Seeed XIAO ESP32-S3/SX1262 Kit	Supports multiple firmwares, SX1262 radio well-established, common and compact XIAO form factor
Power Mgmt.	Waveshare Solar Power Manager (D); 18650 Li-ion	Forward compatible, enables > 6 h runtime with three 2500 mAh cells.
USB Hub	Waveshare USB Hub HAT	Solderless integration via pogo pins to SBC test pads; expands USB and Ethernet.

5.2.3 Enclosure Design and Materials

The enclosure is a custom-designed, 3D-printed housing optimized for thermal isolation.

- **Tiered Isolation:** The enclosure features three internal compartments. The bottom tier houses the battery, power management board, and LTE modem; the middle tier houses the RPi, USB Hub and HaLow module; and the top tier houses the LoRa radio and antennas. This separation prevents the heat load of the LTE modem from causing thermal throttling of the RPi CPU.

5.2.4 Power Management

The payload is powered by up to 3 3.7 V 18650 Li-ion cells in parallel, regulated to 5 V via the Waveshare Solar Power Board (D). A battery holder is connected to the power board. Charging and discharging occurs through the USB-C port of the power board; the board also supports solar charging and charge/discharge via screw terminals.

- **HaLow FEM Power Supply:** The Quectel FGH100M-H module’s High-Power Front-End Module (FEM) is supplied directly from the 5 V main rail of the SBC. This avoids overloading the SBC’s internal 3.3 V LDO regulator, which is rated for lower peak currents.

5.3 Software Stack and Network Topology

The HAP operates as a transparent network gateway, utilizing a layered software approach to manage diverse radio protocols.

5.3.1 Operating System and Orchestration

The system runs **Raspberry Pi OS (Debian Trixie)**. **NetworkManager** handles standard Ethernet, 2.4GHz WiFi, and LTE (CDC-ECM) traffic. The HaLow network is managed by a custom script using the Sub-1GHz tooling provided by Morse Micro. Services and routines for the communications stack are managed through **systemd** services.

5.3.2 Morse Micro Linux Driver

The HaLow radio is managed by the `morse` SPI driver, which is loaded as a `systemd` service. A critical implementation detail is the use of the **Board Configuration File (BCF)** provided by Quectel. The BCF contains module-specific RF calibration parameters and ensures the module operates within ISM power limits while achieving the target 27 dBm output. If the incorrect BCF is loaded, the driver will default to a failsafe configuration that effectively disables the radio.

5.3.3 HaLow Access Point

The HaLow Access Point is activated with the `systemd` service `halow-ap.service`, which creates the network using the Morse Micro `hostapd.sig` daemon. Firewall configuration allows passthrough of all HaLow network traffic. For more detail, see appendix D.

5.3.4 Reticulum Network Stack

Reticulum serves as the core transport layer for mesh communications.

- **rnsd:** The Reticulum daemon manages the LoRa and HaLow (IP) interfaces. It is configured to automatically enable connected radios, ensuring system start even if a module is disconnected.
- **LXMF (Lightweight eXchange Mesh Format):** Provides the messaging protocol. The HAP acts as a **Propagation Node**, allowing users to send and receive messages asynchronously even without a direct end-to-end link. Voice messaging and file transfer are natively supported within the framework, and the protocol offers audio compression to facilitate voice communications over very-low-bandwidth links.

5.4 Client-Side Implementation

Standard devices (smartphones, laptops) can connect to the HAP using IP or radio modules managed by the Reticulum software on the client device. Existing projects were utilized to demonstrate potential access methods.

5.4.1 HaLow-to-WiFi Bridge

The client device is an **ESP32-S3** paired with a XIAO HaLow backpack. It creates a local 2.4 GHz WiFi hotspot. The original bridge firmware is an open-source project using the Morse Micro ESP32 SDK [8].

- **Functionality:** The bridge encapsulates WiFi traffic into 802.11ah packets for transport to the HAP. When connected to the local 2.4 GHz hotspot, the user has access to the HAP Internet connection via the HaLow link.
- **Diagnostic Endpoint:** The bridge firmware was modified to include a HTTP endpoint that serves real-time RSSI and SNR data, facilitating field testing and link characterization.

5.4.2 LoRa Clients

The HAP is configured for Reticulum LoRa communications.

- **RNode Firmware:** ESP32/SX1262 module programmed with RNode firmware for direct integration with Reticulum on mobile or desktop devices. Both the client and HAP modules are programmed with RNode firmware.
- **Configuration:** HAP and Client interfaces must be set to the same LoRa parameters – frequency, bandwidth, spreading factor, and coding rate must match.

Any LXMF capable application may be used for messaging over the Reticulum stack. A non-exhaustive list of clients and useful Reticulum applications can be found at the `lorien/awesome-reticulum` repository [9].

The primary Android client tested is the Sideband application for voice message and text communications [10].

While not implemented, the HAP could additionally interface with **Meshcore/Meshtastic** networks to act as a repeater node for those mesh network protocols. Adding this feature requires addition of another LoRa module to the HAP as these networks do not use the Reticulum protocol.

6 Final Design Performance Evaluation

This section presents the testing methodology, results, and performance analysis of the final HAP communications prototype. Testing was conducted across two communication protocols (LoRa and Wi-Fi HaLow) and included both indoor integration validation and outdoor range characterization. The results are evaluated against the design specification established in Section 3.

6.1 LoRa Range Testing

The first phase of outdoor testing evaluated the LoRa radio link using a phone-to-phone (LoRa-to-LoRa) configuration across six test locations in Kelowna, BC. Both devices used omnidirectional monopole antennas oriented vertically. The transmitter was located in an apartment at approximately 20 m above ground level (elevation ~ 421 m above sea level (ASL)). Tests were conducted at distances ranging from 95 m to 5.4 km under varying line-of-sight conditions. Signal-to-noise ratio (SNR), received signal strength indicator (RSSI), and link quality were recorded at each test point.

Table 7 summarizes the results across all six locations.

Table 7: LoRa-to-LoRa range test results across six locations in Kelowna, BC

Test #	3D Dist. (m)	SNR (dB)	RSSI (dBm)	Link Quality (%)	Path Condition
1	95	12.0	-82	100	Direct
2	324	7.8	-120	100	Direct
3	1065	3.2	-116	81.7	Trees in way
4	3218	-5.2	-120	0.25	Semi-direct
5	3436	-0.5	-118	56.7	Direct
6	5446	2.2	-117	75	Direct

Table 8 provides the elevation and distance details for each test point, including the 3D distance accounting for terrain elevation differences.

Table 8: Elevation and distance details for LoRa test locations

Test #	2D Dist. (m)	Elev. Diff. (m)	3D Dist. (m)	Elev. A (m ASL)	Elev. B (m ASL)	Apt. Offset Applied
1	92.51	21.62	95.01	420.91	399.29	Yes
2	323.57	11.01	323.75	420.91	409.90	Yes
3	1064.70	5.51	1064.71	420.91	426.43	Yes
4	3207.61	254.87	3217.72	420.91	675.78	Yes
5	3422.30	307.92	3436.12	420.91	728.83	Yes
6	5436.88	307.45	5445.56	728.81	421.36	No

6.2 Signal Strength Analysis

Figure 3 shows the relationship between SNR and distance across all six test points. The general trend follows the expected inverse relationship, with SNR decreasing as distance increases. A notable anomaly occurs at test points 4 (3.2 km), where the SNR drops sharply to -5.2 dB despite being at a similar distance to test point 5. This is attributed to a semi-direct path with line-of-sight obstructions. Since both devices used omnidirectional monopole antennas pointing vertically, horizontal communication relied on the antennas' sidelobe radiation patterns, making the link particularly sensitive to terrain obstructions.

Test point 6 at 5.4 km recorded a positive SNR of 2.2 dB with 75% link quality, demonstrating that reliable LoRa communication is achievable at the 5 km design target (R11) when direct

line-of-sight is available.

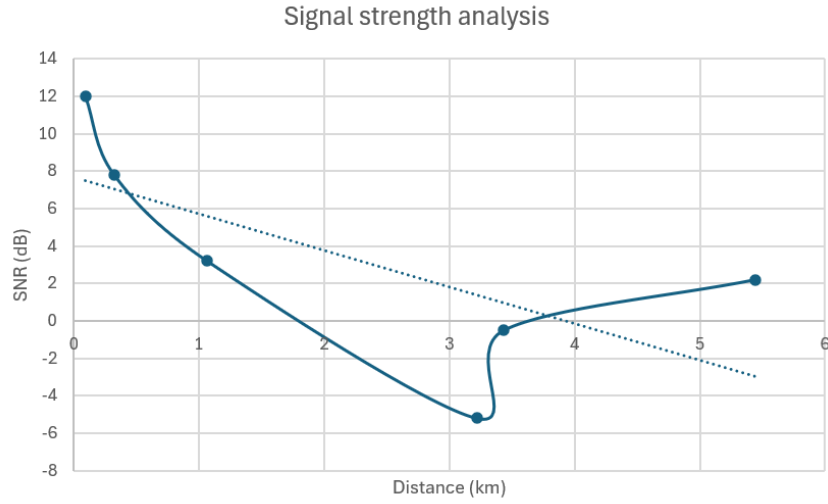


Figure 3: SNR versus distance for LoRa-to-LoRa range testing. The dotted line indicates the general signal decay trend. The dip at approximately 3.2 km corresponds to a test point with an obstructed line-of-sight.

6.3 Elevation profile and Signal Quality

Figure 4 presents a combined view of signal health and terrain elevation. The upper panel overlays SNR and RSSI as functions of distance, showing both metrics degrading with range. The lower panel maps each test point onto an elevation profile with SNR encoded as marker colour intensity. Test points 4 and 5, located at elevations of 675 m and 729 m ASL (approximately 255 m to 308 m above the transmitter), show the effect of significant altitude difference on signal propagation. The elevation variation in these tests partially simulates the altitude differential expected in a true high-altitude deployment.

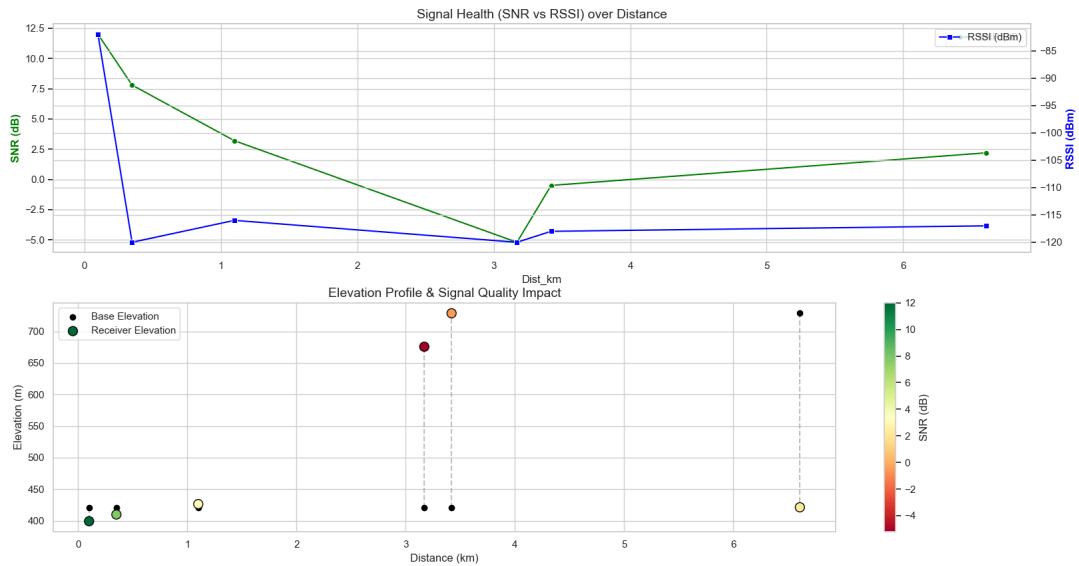


Figure 4: Signal health and elevation profile analysis. Top: SNR (green) and RSSI (blue) versus distance. Bottom: Elevation profile with SNR colour-coded at each receiver location. Base elevation (black dots) and receiver elevation (colored dots) are shown, with dashed lines connecting paired test points at approximately 3 km.

6.4 Geographic Test Coverage

Figure 5 presents all test locations on a satellite map of Kelowna, including the six successful LoRa-to-LoRa tests and the failed 11.5 km HAP integration test. Test points colour follows the signal strength legend: green for strong ($> 5\text{ dB}$), yellow for fair ($0 - 5\text{ dB}$), pink for poor ($< 0\text{ dB}$), and red for failed (no link established).

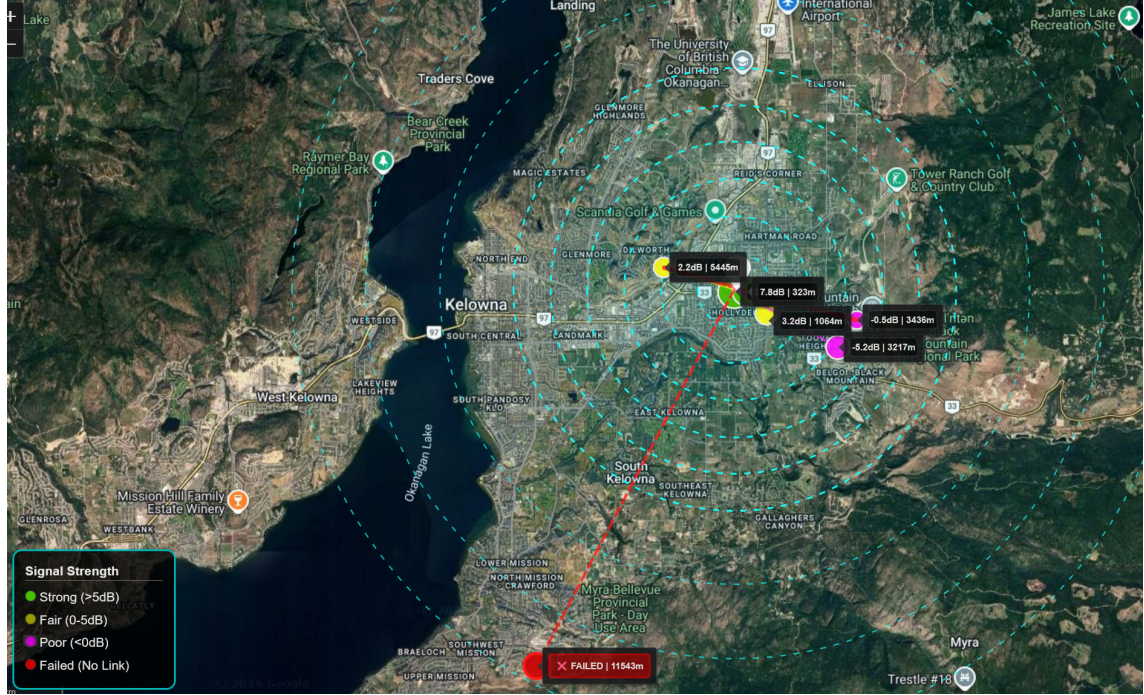


Figure 5: Geographic map of all test locations in Kelowna, BC. Colored markers indicate signal quality at each LoRa-to-LoRa test point, with SNR and distance labels. The red marker at 11,543 m shows the failed LoRa-to-HAP integration test. Dashed cyan circles indicate distance intervals from the transmitter base.

6.5 LoRa-to-HAP integration Test

A long-range integration test was conducted to evaluate the complete LoRa (phone) to LoRa (HAP) communication path at a distance of approximately 11.5 km. The LoRa-to-HAP link was successfully validated in an indoor environment at short range, confirming that the end-to-end communication path between the phone-based LoRa client and the Raspberry Pi-hosted LoRa module functions correctly. However, the link could not be established at the 11.5 km outdoor test distance.

The failure is attributed to a combination of factors: the distance exceeded the demonstrated LoRa-to-LoRa range of 5.4 km; potential configuration mismatches between the phone module and the HAP module (spreading factor, bandwidth, frequency settings); and antenna performance limitations on the HAP side. This result establishes the current system's operational range ceiling between 5.4 km (LoRa-to-LoRa-proven) and 11.5 km (LoRa-to-HAP-failed), and identifies full LoRa-to-Pi integration and RF optimization as key areas for further development.

6.6 Wi-Fi HaLow Testing

Wi-Fi HaLow connectivity between the HAP (Raspberry Pi) and client device was validated through indoor iperf throughput testing. The measured UDP throughput performance over the HaLow link, used to characterize bandwidth capacity and link stability, is summarized in Table 9.

The HaLow RSSI of -61 dBm indicates a strong signal, and the measured throughput of approximately 4.9 Mbps significantly exceeds the $\geq 2\text{ kbps}$ design requirement (R2). Figures 6 and 7 show the iperf output from the client and HAP sides, respectively.

Table 9: Wi-Fi HaLow iperf throughput test results

Parameter	Result
Protocol	UDP
Data Transferred	7 MBytes
Test Duration	12,086 ms
Average Bandwidth (Client)	4,929 kbps (≈ 4.9 Mbps)
Throughput Range (HAP)	4.13 – 7.40 Mbits/sec
HaLow RSSI	−61 dBm

```
Iperf Report
Remote Address: 192.168.60.1:49151
Local Address: 0.0.0.0:5001
Transferred: 7 MBytes, duration: 12086 ms, bandwidth: 4929 kbps
Waiting for client to connect...
HaLow RSSI: -61 dBm
```

Figure 6: HaLow iperf test results from the client side, showing 7 MB transferred at 4,929 kbps bandwidth with an RSSI of −61 dBm.

```
right@pi:~$ iperf -c 192.168.60.235 -p 5001 -i 1 -u -b 10M
Client connecting to 192.168.60.235, UDP port 5001
Sending 1470 byte datagrams, IPG target: 0.00 us (kalman adjust)
UDP buffer size: 280 KByte (default)

[ 1] local 192.168.60.1 port 49151 connected with 192.168.60.235 port 5001
[ ID] Interval      Transfer      Bandwidth
[ 1] 0.0000-1.0000 sec  623 Kbytes  5.19 Mbits/sec
[ 1] 1.0000-2.0000 sec  528 Kbytes  4.33 Mbits/sec
[ 1] 2.0000-3.0000 sec  762 Kbytes  5.75 Mbits/sec
[ 1] 3.0000-4.0000 sec  818 Kbytes  6.78 Mbits/sec
[ 1] 4.0000-5.0000 sec  854 Kbytes  7.00 Mbits/sec
[ 1] 5.0000-6.0000 sec  761 Kbytes  6.13 Mbits/sec
[ 1] 6.0000-7.0000 sec  831 Kbytes  6.81 Mbits/sec
```

Figure 7: HaLow iperf test results from the HAP (Raspberry Pi) side, showing per-interval UDP throughput ranging from 4.13 to 7.40 Mbits/sec.

6.7 LTE Backhaul and End-to-End internet Validation

LTE internet backhaul was implemented and tested on the HAP. During development, the full end-to-end communication chain was validated: Client (iPhone 15 Pro Max) → Wi-Fi HaLow → HAP (Raspberry Pi) → LTE → Internet. A speed test was conducted through this chain using the Ookla Speedtest application, with results shown in Figure 8 and summarized in Table 10.

Table 10: End-to-end internet speed test results (Client → HaLow → HAP → LTE → Internet)

Parameter	Result
Download Speed	0.39 Mbps
Upload Speed	0.45 Mbps
Ping	106 ms
Jitter	78 ms
Network Provider	Rogers Wireless
Client Device	iPhone 15 Pro Max

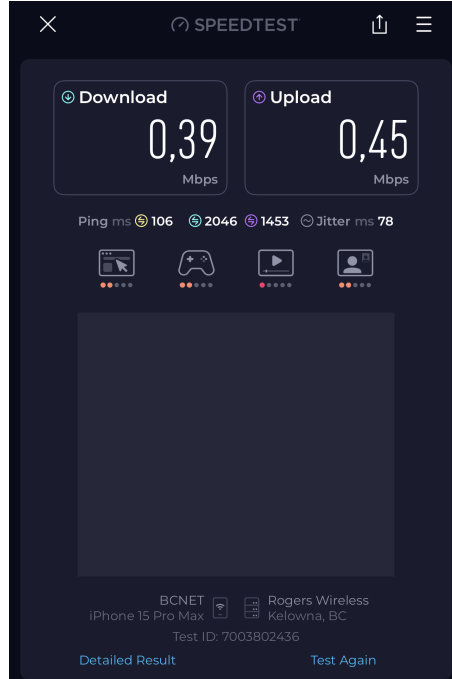


Figure 8: Ookla Speedtest result conducted through the full HAPS communication chain: Client (phone) → Wi-Fi HaLow → HAP → LTE → Internet. Download speed of 0.39 Mbps and upload speed of 0.45 Mbps were achieved via Rogers Wireless in Kelowna, BC.

While the measured speeds are modest compared to direct cellular connections, they are sufficient for the intended emergency communication use case, which requires text messaging and low-rate data transfer. The download speed of 0.39 Mbps (≈ 390 kbps) far exceeds the ≥ 2 kbps design requirement (R2). The higher latency (106 ms ping, 78 ms jitter) is expected given the multi-hop nature of the communication chain. This result validates the core HAPS use case: providing internet connectivity to ground users via an elevated relay platform built with off-the-shelf components.

6.8 Performance Summary

Table 11 provides an overview of tested performance across all communication protocols.

Table 11: Communication protocol performance summary

Protocol	Max Range Achieved	Throughput	Link Quality	Test Type
LoRa (phone-to-phone)	5.4 km	2.73 kbps	75% at max range	Outdoor
LoRa (phone-to-HAP)	Indoor only	24.88 kbps	Functional	Indoor
Wi-Fi HaLow	Indoor only	~4.9 Mbps	Strong (−61 dBm)	Indoor
LTE backhaul	N/A	0.39 / 0.45 Mbps	Functional (106 ms ping)	Indoor

7 Project Risk Assessment

The HAPS Project involves many technical, operational, and ethical risks that need to be acknowledged and mitigated. The design specifications focus on maximizing the communication range by using LoRa for low-bandwidth emergency communication, providing significant coverage for rural and remote regions [2], [11].

7.1 Project Risk Identification and Characterization

7.1.1 Technical Risks

Table 12: Technical Risks

Risk	Description	Prob.	Sev.	Mitigation Strategy
Insufficient Range	Range > 2km limited by alignment or stability [12].	5/10	High	Link budget simulation; LoRa modulation optimization [13].
Power Runtime	Limited battery capacity for multi-day mission support.	5/10	Med	Duty cycle scheduling; low-power LoRa modules.
Payload Mass	Lift capacity decreases with mass [12].	3/10	High	Ensure integrated payload remains < 2 kg.
Weather Instability	Wind/rain causing damage or loss of line-of-sight.	8/10	Med	Tethered flight protocol; testing during calm-weather windows.

7.1.2 Project Management and Safety Risks

Table 13: Management and Safety Risks

Risk	Description	Prob.	Sev.	Mitigation Strategy
Schedule Delays	Simulation/Integration complexity [14].	5/10	Med	WBS tracking; weekly internal deliverables.
Budget Overrun	Costs exceeding the \$500 CAD limit.	5/10	Med	Real-time cost tracking; part upcycling.
Balloon Safety	Enclosure collapse or tether failure [15].	5/10	Med	Strict PPE; safe perimeter protocols.

7.2 Risk Matrix and Impact Analysis

The following matrix visualizes the intersection of probability and severity for the identified project risks.

Table 14: Risk Matrix Visualization

Sev./Prob.	Low	Medium	High
High	Safety injury	Range and link failure	Severe weather instability
Medium	Field logistics	Power/runtime limits	Gateway software crash
Low	Cost fluctuations	Component availability	N/A

7.3 Failure Mode and Effects Analysis (FMEA)

To prioritize specific technical risks of the final prototype, an FMEA was performed. Each failure is assigned a Risk Priority Number ($RPN = Severity(S) \times Occurrence(O) \times Detection(D)$). Severity determines the impact of the failure if it were to occur; occurrence is the likelihood of it occurring; and detection is linked to how well we can catch the failure. If the $RPN \geq 100$ it indicates a high-priority failure which requires technical mitigation.

Table 15: System-Level Failure Mode and Effects Analysis (FMEA)

Component	Failure Mode	S	O	D	RPN	Recommended Action / Mitigation
Enclosure	Thermal buildup (Solar radiation + Device self heat)	7	6	5	210	Use "BOB V1.0" layered isolation; separate LTE module from Raspberry Pi
Gateway	Software hang due to multi-protocol overhead	8	5	4	160	Deploy Reticulum network stack
Power Unit	4G/LTE current spikes causing system brownout	8	6	3	144	Use high-discharge 18650 cells and decoupled power rails
Software	USB Bus bandwidth limitation between modules	6	4	5	120	Offload LoRa processing to ESP32 MCUs to reduce USB overhead
Link	Signal loss due to uncontrolled balloon movement in deployment	6	5	2	60	Use omnidirectional dipole antennas

7.4 Technical Mitigation and Justification

The following strategies address the highest RPN scores ($RPN \geq 100$) to ensure the system is reliable in deployment.

7.4.1 Thermal Management and Tiered Isolation (RPN 210)

Thermal management is crucial because traditional cooling is ineffective in low-density, high-altitude environments. The SIM7670G 4G module generates approximately 537 mW of heat, which can cause the Raspberry Pi Zero 2W to reach its thermal throttling limit.

Mitigation: The design utilizes a 3-tiered enclosure to physically isolate the heat-generating LTE radio from the Raspberry Pi and other component modules.

Justification: This layered isolation maintains the CPU temperature below its 80 °C thermal threshold, preventing software and hardware lag, ensuring the bi-directional link remains usable.

7.4.2 Network Resilience via Reticulum (RPN 160)

Managing LoRa, WiFi HaLow, and 4G LTE data links creates a very high complexity ceiling, as the drivers can conflict, leading to the system lagging or, in the worst case, stopping transmission and reception.

Mitigation: Integration of the Reticulum network stack such that the software can communicate between the modules without caring which specific radio hardware is being used.

Justification: Reticulum abstracts the hardware layer. If the LTE interface fails, the stack allows rerouting over the LoRa link, preventing a total communication blackout.

7.4.3 Power Stability and Surge Handling (RPN 144)

4G/LTE modules require high transient current during operation (transmitting and receiving), which can drop the system's voltage below the 4.7 V needed for operation.

Mitigation: Make use of high-discharge 18650 cells and decoupled power rails for radio modules and Raspberry Pi.

Justification: Isolating the Raspberry Pi from transient surges generated by the radio hardware prevents brownouts of the Raspberry Pi, thus ensuring runtime is preserved, and non-intentional blackouts are avoided.

7.5 Summary of Risk Mitigation

Extended range specifications introduce manageable risks that are effectively addressed through system optimization, in-depth analysis of power-efficiency, and staged prototype testing [2], [14]. The benefits of the system outweigh the technical risks; that being said, the HAPS provides a strong communication link during disasters, protecting human life and infrastructure.

8 Ethical Considerations of Final Design

The design and deployment of this project, like any other engineering project, entail several ethical considerations regarding safety, environmental impact, privacy, and professional responsibility. These considerations are based on the Engineers and Geoscientists British Columbia (EGBC) Code of Ethics, which provides the guidelines for how engineers can protect public welfare and uphold their responsibilities towards the engineering profession.

8.1 Safety and Public Welfare

The EGBC Code of Ethics requires engineers to "hold paramount the safety, health, and welfare of the public" [16]. Although this project does not focus on mechanical deployment, nor does it involve high-risk operations at this stage, future integrations and improvements, such as integration with the balloon-based platform, can introduce risks (e.g., payload falling or loss of control due to environmental conditions).

These risks must be mitigated for the system to be safely deployed. Within the current scope of the project, the system is designed with lightweight development components for low-altitude testing under conservative operating conditions. These decisions minimize harm; however, they are also a byproduct of the project's time and complexity constraints. If this project were further developed and more refined hardware and networking capabilities were used to work towards a commercially viable product, strict adherence to public safety must still be ensured despite increasing complexity and risk. This could translate to measures such as compliance with aviation regulations and formal risk assessments to evaluate the public risk associated with this project.

8.2 Environmental Impact

Being a product intended for deployments in remote areas, a key consideration is the environmental impact of deploying such a HAPS system. This is critical, as environmental responsibility is another core component of ethical engineering practice under EGBC guidelines [16]. Consideration must include not only the infrastructure required for deployment but also network operations, the devices used, and end-of-life handling of components. Being an airborne device, potential environmental impacts could include bird strikes, forest clearances for balloon deployments, wildlife interactions, forest fires, and other natural or man-made hazards.

At the current stage, the environmental impact has not yet been studied. However, in the future, the design should be tested for robustness under environmental conditions, more environmentally friendly materials should be explored, and protocols should be established for failed deployments, disassembly, and disposal at the product's end of life.

8.3 User Privacy and Data Security

As a communication system, this project is responsible for protecting user data and preventing unauthorized access or interception. This extends the principle of public welfare and safety, as critical communication could be exploited by bad actors if the system is compromised. Such misuse through data theft, manipulation, or even mass surveillance could directly harm the safety and privacy of the network's users, making data security one of the most important ethical considerations.

Even at this proof-of-concept stage, the system incorporates higher-layer secure protocols such as Reticulum, which provide encryption and secure routing. These design decisions ensure that sensitive information, particularly in emergency communication scenarios, is properly protected. Future development should continue to prioritize stronger encryption, robust user authentication mechanisms, and secure key management to maintain the network's security.

Despite these security measures, future development could still introduce risks such as first-party data collection or unauthorized surveillance. To address this, the design process emphasizes transparency, regulatory compliance, and clearly defined use cases. Ethical engineering practice requires anticipating these potential risks and ensuring that deployment aligns with both legal requirements and socially beneficial applications.

8.4 Regulatory Compliance

The EGBC Code of Ethics requires engineers to act with honesty, competence, and in adherence to applicable laws and regulations [16]. This is particularly relevant in this project, as both LoRa and Wi-Fi HaLow operate in the shared 915 MHz license-exempt spectrum bands.

The current system complies with ISED regulations by limiting transmission power and using standardized communication protocols such as LoRa and Wi-Fi HaLow. These choices ensure that devices within the network use a fair portion of the shared spectrum and avoid interference with other systems operating in the same band. This consideration remains important even if future improvements introduce more advanced or custom protocols.

8.5 Equity and Accessibility

The target use case of this project is to improve access to communication infrastructure in underserved or remote regions for emergency applications, which aligns with the ethical principle of fairness and equity [16]. The current design is also highly accessible due to its modularity, reliance on off-the-shelf components, and minimal integration, which promote ease of use. Future improvements should leverage this modularity to maintain and enhance accessibility as it scales up.

However, ethical responsibility also requires acknowledging the system’s limitations. The HAPS system is designed purely as a supplementary solution rather than a replacement for permanent infrastructure. Ensuring that such technologies do not create long-term dependency is an important aspect of responsible engineering design. For real-world deployments and future iterations, appropriate consultation with stakeholders is essential to identify and address accessibility considerations that might otherwise be overlooked.

9 Project Success Analysis

This section evaluates the final prototype against the design requirements and evaluation criteria established in Section 3.

9.1 Design Requirements Evaluation

The prototype was developed in accordance with the design requirements (R1 to R11) and evaluation criteria (E1 to E4) established in Section 3.

Table 16: Design requirements evaluation against R1-R11

ID	Requirement	Target	Status	Evidence
R1	Bi-directional link	Two-way wireless link	Met	LoRa and HaLow both validated for two-way communication
R2	Min. data rate	≥ 2 kbps	Met	HaLow: $\sim 4,929$ kbps (Table 9)
R3	Battery runtime	≥ 30 min	~ 5 h	Battery runtime with 3 cells, LTE, HaLow, LoRa active
R4	Demo platform	Low-altitude	Met	Apartment rooftop (~ 20 m)
R5	Payload mass	< 2 kg	591 g	Weighed assembled prototype
R6	Cost	$\leq \$500$ BOM	Met	Final BOM \$375.14
R7	Regulatory	ISED compliant	Met	Operating in ISM band per RSS-247
R8	Scope compliance	Within agreed scope	Met	No out-of-scope features pursued
R9	Balloon-based HAP	Tethered balloon	Partially Met	Enclosure designed for balloon; demo used rooftop
R10	Regulatory band	ISED licence-exempt	Met	915 MHz ISM band
R11	Extended range	≥ 5 km LOS	Met	LoRa: 5.4 km at 75% link quality (Table 7), text and voice communications

9.2 Evaluation Criteria Assessment

Table 17 assesses the prototype against the four evaluation criteria defined in Section 3, Table 2.

Table 17: Evaluation criteria assessment

Criterion	Assessment	Discussion
E1: Payload mass & power draw	Met	Total payload mass is 0.6 kg. The power system provides 300 min of continuous runtime. Potential for long-distance communications at high data rates is balanced with increased mass and power draw. Modular design allows removal of unneeded radios and lower battery cell count to reduce weight and power draw.
E2: Communication range	Met	LoRa-to-LoRa demonstrated 5.4 km, meeting the ≥ 5 km target (R11). Full LoRa-to-HAP link validated indoors only; the 11.5 km outdoor test failed. HaLow tested indoors only.
E3: Reliability & availability	Partially Met	LoRa achieved 75-100% link quality up to 5.4 km with direct LOS. Link quality dropped to $< 1\%$ with obstructed paths. HaLow showed stable throughput at an RSSI of -61 dBm indoors. Long-term reliability was not tested.
E4: Sustainability	Met	Modular, reusable off-the-shelf components. A 3D-printed enclosure can be reprinted or modified. ASA filament is recommended for UV-resistant deployment versions.

9.3 Key Achievements

The project achieved the following:

- Bi-directional LoRa communication validated at 5.4 km, meeting the ≥ 5 km design target (R11).
- Wi-Fi HaLow throughput of ~ 4.9 Mbps, exceeding the 2 kbps data rate target (R2) by over $2,000\times$.
- End-to-end internet chain (Client $\rightarrow$ HaLow $\rightarrow$ HAP $\rightarrow$ LTE $\rightarrow$ Internet) validated with measured speeds of 0.39 Mbps download and 0.45 Mbps upload, confirming the core HAPS relay use case.
- LTE internet backhaul successfully integrated into the HAP.
- A modular 3D-printed enclosure designed, built, and physically tested.
- System remained within the \$500 BOM constraint.

9.4 Limitations

Several limitations were identified during testing:

- LoRa-to-HAP integration was validated indoors but failed at 11.5 km outdoor range, indicating further RF optimization and configuration tuning is needed.
- Wi-Fi HaLow and LTE testing were limited to indoor environments. Outdoor range characterization was not completed due to time and equipment constraints.
- All tests were ground-based. No high-altitude deployment testing was conducted; rooftop elevation (~ 20 m) served as a proxy.
- The enclosure lacks environmental sealing for protection against moisture, dust, and temperature.

9.5 Overall Assessment

The HAPS communications prototype meets the primary project's objective of demonstrating a functional, bi-directional, multi-protocol communication system suitable for high-altitude platform deployment. The LoRa subsystem met the ≥ 5 km range target (R11) with a demonstrated range of 5.4 km, the HaLow subsystem demonstrated throughput far exceeding the 2 kbps requirement (R2), and LTE integration provides a viable internet backhaul path validated through end-to-end speed testing. While full system integration at range remains incomplete, the individual subsystem validation confirms the technical feasibility of the design approach and provides a solid foundation for future development.

References

- [1] International Telecommunication Union, “Measuring digital development : Facts and figures 2024,” Measuring Digital Development, 32 p. 2024, ISSN: 978-92-61-39861-3. [Online]. Available: <http://digitallibrary.un.org/record/4074377>.
- [2] S. C. Arum, D. Grace, and P. D. Mitchell, “A review of wireless communication using high-altitude platforms for extended coverage and capacity,” *Computer Communications*, vol. 157, pp. 232–256, May 1, 2020, ISSN: 01403664. DOI: 10.1016/j.comcom.2020.04.020. Accessed: Oct. 27, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0140366419313143>.
- [3] Y. Zeng, Q. Wu, and R. Zhang, “Accessing from the sky: A tutorial on UAV communications for 5g and beyond,” *Proceedings of the IEEE*, vol. 107, no. 12, pp. 2327–2375, Dec. 2019, ISSN: 1558-2256. DOI: 10.1109/JPROC.2019.2952892. Accessed: Nov. 1, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/8918497>.
- [4] H. Soy, “Coverage analysis of LoRa and NB-IoT technologies on LPWAN-based agricultural vehicle tracking application,” *Sensors (Basel, Switzerland)*, vol. 23, no. 21, p. 8859, Oct. 31, 2023, ISSN: 1424-8220. DOI: 10.3390/s23218859. Accessed: Oct. 27, 2025. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10648585/>.
- [5] M. Gemignani and S. Marcuccio, “Dynamic characterization of a high-altitude balloon during a flight campaign for the detection of ISM radio background in the stratosphere,” *Aerospace*, vol. 8, no. 1, p. 21, Jan. 2021, ISSN: 2226-4310. DOI: 10.3390/aerospace8010021. Accessed: Oct. 27, 2025. [Online]. Available: <https://www.mdpi.com/2226-4310/8/1/21>.
- [6] Government of Canada. “RSS-247 — digital transmission systems, frequency hopping systems and licence-exempt local area network devices in 902-928 MHz, 2400-2483.5 MHz, 5150-5350 MHz, and 5470-5895 MHz bands.” Last Modified: 2025-08-11, Accessed: Dec. 6, 2025. [Online]. Available: <https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en/devices-and-equipment/radio-equipment-standards/radio-standards-specifications-rss/rss-247-digital-transmission-systems-dtss-frequency-hopping-systems-fhss-and-llicence-exempt-local>.
- [7] Morse Micro. “Pushing the limits: Wi-fi HaLow testing in joshua tree national park,” Morse Micro, Accessed: Apr. 1, 2026. [Online]. Available: <https://www.morsemicro.com/2024/09/09/pushing-the-limits-wi-fi-halow-testing-in-joshua-tree-national-park/>.
- [8] gtgreenw, *Gtgreenw/xiao-halow-to-WiFi-bridge*, original-date: 2026-01-31T14:15:30Z, Mar. 10, 2026. Accessed: Apr. 1, 2026. [Online]. Available: <https://github.com/gtgreenw/Xiao-Halow-to-WiFi-Bridge>.
- [9] lorien, *Lorien/awesome-reticulum*, original-date: 2025-12-21T22:25:27Z, Mar. 31, 2026. Accessed: Apr. 1, 2026. [Online]. Available: <https://github.com/lorien/awesome-reticulum>.
- [10] Mark Qvist. “Sideband,” Accessed: Apr. 2, 2026. [Online]. Available: <https://unsigned.io/software/Sideband.html>.
- [11] X. Cao, P. Yang, M. Alzenad, X. Xi, D. Wu, and H. Yanikomeroglu, “Airborne communication networks: A survey,” *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 9, pp. 1907–1926, Sep. 2018, ISSN: 1558-0008. DOI: 10.1109/JSAC.2018.2864423. Accessed: Nov. 3, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/8438489>.
- [12] G. Karabulut Kurt et al., “A vision and framework for the high altitude platform station (HAPS) networks of the future,” *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 729–779, 2021, ISSN: 1553-877X. DOI: 10.1109/COMST.2021.3066905. Accessed: Nov. 1, 2025. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/9380673>.
- [13] A. Fotouhi et al., “Survey on UAV cellular communications: Practical aspects, standardization advancements, regulation, and security challenges,” *IEEE Communications Surveys & Tutorials*, vol. 21, no. 4, pp. 3417–3442, 2019, ISSN: 1553-877X. DOI: 10.1109/COMST.2019.2906228. Accessed: Nov. 3, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/8675384>.
- [14] G. Castellanos, M. Deruyck, L. Martens, and W. Joseph, “Performance evaluation of direct-link backhaul for UAV-aided emergency networks,” *Sensors*, vol. 19, no. 15, p. 3342, Jan. 2019, ISSN: 1424-8220. DOI: 10.3390/s19153342. Accessed: Nov. 3, 2025. [Online]. Available: <https://www.mdpi.com/1424-8220/19/15/3342>.
- [15] A. Kumbhar, S. Singh, and I. Guvenc, “UAV assisted public safety communications with LTE-advanced HetNets and FeICIC,” in *2017 IEEE 28th Annual International Symposium on Personal, Indoor, and Mobile Radio Communications (PIMRC)*, Oct. 2017, pp. 1–7.

- DOI: 10.1109/PIMRC.2017.8292564. Accessed: Nov. 1, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/8292564>.
- [16] Engineers and Geoscientists British Columbia, “Code of ethics,” Accessed: Mar. 25, 2026. [Online]. Available: <https://www.egbc.ca/complaints-discipline/code-of-ethics/code-of-ethics>.
 - [17] Y. Khan, *Ykhan1999/zero2w_80211ah*, original-date: 2025-09-29T15:16:43Z, Mar. 31, 2026. Accessed: Apr. 1, 2026. [Online]. Available: https://github.com/ykhan1999/zero2w_80211ah.
 - [18] Morse Micro, “Linux porting guide,” Morse Micro Pty. Ltd., v2, May 30, 2025. [Online]. Available: https://www.morsemicro.com/resources/appnotes/MM_APPNOTE-24_Linux_Porting_Guide.pdf.
 - [19] Mark Qvist. “Reticulum network stack 1.1.3 documentation,” Accessed: Apr. 2, 2026. [Online]. Available: <https://reticulum.network/manual/index.html>.
 - [20] SIMCom Wireless Solutions Limited, “SIM767xx series_at command manual,” v1.01, Sep. 25, 2023. Accessed: Apr. 1, 2026. [Online]. Available: https://files.waveshare.com/wiki/SIM7670G-LTE-Cat-1-GNSS-HAT/SIM767XX_Series_AT_Command_Manual_V1.01.pdf.
 - [21] ajudge. “SNR vs MCS rate for MM6108 - general,” Morse Micro Community. Section: General, Accessed: Apr. 1, 2026. [Online]. Available: <https://community.morsemicro.com/t/snr-vs-mcs-rate-for-mm6108/1266>.

A Team Member Contributions

Table 18: Member Contribution

Deliverable	Musab	Ahmed	Sirraaj	Abdul	John	Eris
Meeting 1 (Project Scope)	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant
Meeting 2 (Design Specs)	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant
Conceptual Design Report	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant
Conceptual Design Presentation	Slides, Presenter	Slides, Presenter	Slides, Presenter	Slides, Presenter	Slides, Presenter	Slides, Presenter
Meeting 3 (Proposed Solutions)	Doc Sections, Review, Participant	Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant
Meeting 4 (Improvements)	Doc Sections, Review, Participant	Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant
Meeting 5 (Future Iterations)	Doc Sections, Review, Participant	Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant	Doc Sections, Review, Participant
Prototype	Prototype Testing, LoRa Module Testing, Enclosure Improvements, Final Assembly, Minor Parts Procurement, 3D Printing	Prototype Testing, Final Assembly, Wi-Fi HaLow & LoRa Module Testing	Enclosure Design and Development, Final Assembly, Prototype Testing	Prototype Testing, Final Assembly, LTE & LoRa Module Testing	Prototype Testing, LoRa Module Testing	Major Parts Procurement, Major Development, Testing, Final Assembly, Software Development, Firmware
Poster	Design & Content	Review	Figure creation & Review	Figure creation & Review	Review	Review
Video Presentation	Filmed Presentation, Slide Section, Review	Filmed Results, Slide Section and Review	Filmed Results, Editor, Slide Section, Review	Filmed Results, Slide Section, Review	Slide Section, Review	Filmed Results, Slide Section, Review
Final Design Report	Doc Sections, Figures, Review	Doc Sections, Review	Doc Sections, Review	Doc Sections, Review	Doc Sections, Review	Doc Sections, Review

B Financial Summary

Table 19: Budgeted Cost Breakdown for Design Solution 1&2 from Meeting 3

Item/Part	Description	Unit Cost (CAD)	Qty	Total (CAD)
SX1278	LoRa radio transceiver	\$8.50	2	\$17.00
Raspberry Pi Pico 2W	HAP microcontroller	\$23.03	2	\$46.60
6 Pack 915MHz Antenna	5 dBi antenna	\$15.97	1	\$15.97
Lithium Ion Battery	3.7V 2000mAh	\$11.95	1	\$11.95
Power Management	Adafruit bq25185 (charging / power management)	\$11.95	1	\$11.95
Solar Panel	Small solar panel (trickle charging)	\$11.99	1	\$11.99
Thermistor	Battery temperature safety sensor	\$9.18	1	\$9.18
USB-to-LoRa Dongle	Ground node link dongle	\$33.99	1	\$33.99
Implementation Costs	Miscellaneous (wires, solder, connectors, etc.)	\$50.00	1	\$50.00
Total Design Solution Cost				\$208.63

Table 20: Real Cost of Purchased Components

Item	Quantity	Procurer	Final w/ Tax
915MHz LoRa Antenna (4x)	1	Musab	22.38
Ethernet/USB HUB HAT	1	Musab	24.72
Raspberry Pi Zero 2 WH - with Pre-Soldered Headers	1	Eris	44.79
Waveshare Solar Power Manager (D)	1	Eris	20.79
XIAO ESP32S3 & Wio-SX1262 Kit w/ 3D Case for Mesh-tastic & LoRa	2	Eris	61.02
Seeed Studio XIAO ESP32S3 (3x)	1	Eris	33.08
SanDisk Extreme® microSDXC™ UHS-I card, 128GB	1	Eris	33.91
Female SMA Antenna Connector (20x)	1	Eris	15.33
2PCS/Lot LoRa2021	1	Eris	23.79
Wi-Fi Halow Transceiver Module Board for Seeed Studio XIAO, 915MHz 902-928MHz, FGH100M-H	3	Eris	63.54
SIM7670G, w/LTE and GPS antenna	1	Eris	31.79
Total Ordered Parts Cost			\$375.14

C HaLow Integration: Kernel Compilation, Driver Patching, MCU Firmware

The 802.11ah link using the MM6108 radio requires a custom Linux kernel environment, as the Morse Micro driver is not yet in the mainline kernel.

1. **Kernel Source:** Utilize the Morse Micro Linux kernel fork in the `MorseMicro/rpi-linux` repository or patch the current Raspberry Pi OS kernel (Debian Trixie) using the patches provided in the `morse.driver` repository. The Morse Micro forums contain guides on implementing the Linux driver; there are stated plans from Morse Micro to produce a guide on applying the relevant patches to more recent kernel versions.
 - For simplicity, the prototype uses the repo at `ykhan1999/zero2w.80211ah` [17] as a base, loosely following instructions for setup without using the quick-start image. This provides the patched kernel, drivers, and Morse Micro tooling for administering 802.11ah networks. Note that not all of the repository provided scripts are used, and modifications are required to adapt for the Seeed HaLow module.
2. **Porting Guide:** Detailed instructions for Linux kernel compilation and driver integration are documented in **MM APPNOTE-24: Linux Porting Guide**, provided by Morse Micro [18].
3. **Modules:** Loading of `morse` module is managed via the `systemd` unit located at `/etc/systemd/system/morse-driver.service`. Configuration of the operational parameters are required at time of loading, and specified in this file.
4. **Firmware (BCF):** The Board Configuration File for the Quectel FGH100M-H is required to calibrate the high-power FEM and must match the Morse Micro firmware version (v1.15+ used for this prototype). Morse Micro maintains a repository of BCF files in the `MorseMicro/morse-firmware` repository. Specifically, the FGH100M-H uses the `bcf_fgh100mhaamd.bin` BCF, which was provided by Quectel for this module.
 - This BCF file should also be used in projects that use newer versions of the `MorseMicro/mm-iot-esp32` SDK for ESP32 MCU projects. At the time of writing, the Seeed fork of the SDK (found in `Seeed-Studio/mm-iot-esp32`) is based on an MM-IoT SDK version 2.6.4, which uses a BCF that is incompatible with later Morse Micro Firmware. Firmware versions are sometimes updated with new MM-IoT SDK versions, so forward compatibility is not necessarily guaranteed.

C.1 Hardware Pinouts and Interface Tables

The connection between the Raspberry Pi Zero 2 W and the Seeed XIAO HaLow module uses the following SPI pinout, adapted to this module from the `ykhan1999/zero2w.80211ah` repo:

Function	RPi GPIO / Pin	XIAO Pin
MOSI	GPIO 10	D10
MISO	GPIO 9	D9
SCLK	GPIO 11	D8
CS	GPIO 8	D3
IRQ	GPIO 25	D2
RESET	GPIO 5	D0
WAKE	GPIO 3	D1
BUSY	GPIO 7	D4
5V Power	5V (Pin 2 or 4)	5V
3.3V Power	3.3V (Pin 1 or 17)	3V3
Ground	GND (Pin 6, 9, 14, etc.)	GND

Table 21: SPI Interface and Power Pinout

D Raspberry Pi Configuration and Setup

This section provides a brief overview of the configuration for the Linux OS and networking stacks. The default login user is `f1` and the password is `f1haps`.

D.1 Software Configuration Files

The Reticulum manual should serve as the primary full reference for configuring interfaces [19] – a summary of specific implementation points for the prototype is given below.

The Reticulum network stack is python based, installed via `pipx` and loaded via `systemd` services. The configuration files are found in `/home/f1/.reticulum/` and `/home/f1/.lxmd/` for `rnsd` and `lxmd` respectively.

- `rnsd` is executed as a system service, with a check to enable only the radio devices connected to the system. This means that a malfunction or absence of either the LoRa or HaLow radios will not prevent the service from starting. `rnsd` provides the Reticulum service for network traffic routing, and `lxmd` provides the LXMF router and propagation node for the messaging protocol.
- e.g. from the terminal, `rnsd` can be started with `sudo systemctl start rnsd` and `lxmd` is started with `systemctl --user start lxmd`
- `~/.reticulum/config.template` contains the template configuration file for `rnsd`; modify this configuration file as the primary file. The `systemd` service script runs a check that will remove the LoRa or HaLow interfaces if absent before starting the service, else the service will fail to start if not all declared devices are present.
- `~/.lxmd/config` contains settings for the LXMF daemon and propagation server configuration.

D.2 HaLow AP

The HaLow AP is managed using special Sub-1GHz utilities provided by Morse Micro. Scripts for loading the `morse` driver and bringing up the HaLow interface are managed by `systemd`:

- `sudo systemctl start halow-ap` starts the HaLow AP.
- `sudo systemctl start start_morse` loads the `morse` kernel module.

These cover the main differences between standard WiFi management and the HaLow management; the `wlan1` device is not managed by `NetworkManager` like other network interfaces as the Sub-1GHz utilities are not natively integrated. The Morse utilities function by presenting the HaLow interface as if it were a 5GHz WiFi interface w.r.t standard Linux utilities.

D.3 LTE Modem Configuration

The SIM7670G modem is configured for CDC-ECM to provide Ethernet over USB. The configuration can be changed as required using AT commands detailed in the SimCom AT command manual [20].

- **APN:** Set to the required APN for the chosen network via AT commands.

Alternatively, the modem can be managed by `ModemManager` if desired, including use of the PPP interface. The module includes GPS functionality; this is untested in the prototype.

E Mobile Telemetry Automation Strategy

This addition proposes a framework for automation of performance testing of the RF links. While this was not implemented for the primary data collection in the report, use of such a framework would improve the collection of testing data for future projects.

To accurately profile the real-world line-of-sight (LOS) performance of both the 802.11ah (HaLow) and LoRa radio links, a mobile telemetry collection system was developed. The strategy relied on

utilizing an Android smartphone as an automated, portable testing client, continuously logging geospatial and RF hardware metrics while traversing the test area.

E.1 Environment Setup and Utilities

The automation environment was hosted using **Termux**, a terminal emulator that provides a Linux environment on Android without requiring root access. To facilitate hardware integration and network testing, several key utilities and companion applications were employed:

- **Termux & Termux:API:** While Termux provided the execution environment for the Python automation script, the **Termux:API** Android companion application is also required. Installed alongside its corresponding command-line package, this companion app bridges the Android security sandbox. It allowed the automation script to query the device’s hardware GPS via the `termux-location` command, accurately geotagging every telemetry reading.
- **Sideband:** This Android application served as the host daemon for the Reticulum Network Stack. It managed the physical connection to the LoRa RNode hardware and exposed a local Shared Instance for the Termux environment to utilize.
- **iperf3:** Utilized for active TCP bandwidth testing over the 802.11ah HaLow link.
- **Reticulum Network Stack (rns):** The `rnprobe` utility was leveraged to actively ping the remote propagation node’s dedicated transport responder, allowing the script to extract LoRa hardware metrics (RSSI, SNR, and RTT).
- **HaLow Bridge Endpoint:** A http endpoint was implemented in the firmware of the HaLow module, providing link RSSI as reported by the Morse Micro SDK utilities, and SNR derived based on the specified channel bandwidth using the methodology provided via the Morse Micro forum [21].

E.2 Data Collection Methodology

The core of the automation was a Python script designed to operate on a continuous 10-second heartbeat loop. During each iteration, the script executed the following sequence to capture a complete snapshot of the network state:

1. **Geospatial Tagging:** Polled the Termux:API to retrieve the most recent device coordinates (latitude and longitude).
2. **HaLow Polling:** Sent an HTTP GET request to the ESP32 endpoint to log the 802.11ah RSSI, SNR, and channel bandwidth, immediately followed by a brief `iperf3` throughput test.
3. **LoRa Probing:** Issued an `rnprobe` request through the Sideband shared instance to the remote gateway. The script parsed the cryptographic reply to record the LoRa RSSI, SNR, and round-trip time.
4. **Data Aggregation:** Appended the aggregated timestamps, location data, and dual-band metrics to a local CSV file for post-test geospatial plotting.

E.3 Fault Tolerance and OS Constraints

Given the volatile nature of edge-of-coverage RF testing, the automation was optimized for fault tolerance. Strict asynchronous timeouts were enforced on all network calls (HTTP requests, `iperf3` execution, and Reticulum probes) to prevent the script from halting during periods of complete signal loss or asymmetric link states. System wake-locks were engaged within Termux to ensure the OS did not disrupt the script as it runs.